Why the multi-target IC stays above BSGS (option 3, 2026-11-11)

Question: the multi-target IC (IC4, `research/mt_fieldops_v2_20261111/`) is 1.18x to 1.80x a paired BSGS per target (1.49x to 2.04x without the shared verification constant; claim PCC-20261111-fa13c8 [pcc:unverified]). Can the IC close that gap?

The mechanism

Both methods build a table of K x-classes at the same cost per class (one shared inversion per two classes) and then try one point per step. They differ in which logs the table covers:

- **IC:** the table logs are sums of factor-base logs, i.e. effectively random. A trial point Q + w is in the table with probability 2K/n, independently of earlier trials, so the expected number of trials is n/(2K).
- **BSGS:** the baby table covers the interval |k| <= Mx and the giant offsets tile the group with disjoint windows, so the hit trial is uniform over the windows: expected trials 1 + n/(4 Mx), half as many.

At the optimum (table cost = expected target cost) a factor 2 in trials becomes a factor sqrt(2) in total cost, before fixed costs. That is the structure of the measured gap.

Check 1: the models predict the measured trial counts (v2 holdout rows, no new run)

`trials_check.py` on `research/mt_fieldops_v2_20261111/results/holdout/rows.jsonl` (d = fraction of factor-base logs determined; a hit is usable with probability about d^2):

bits	T	ic4 measured / n/(2Kd^2)	bsgs2 measured / (1 + n/(2(2Mx+1)))
22	1024	1.011 (22 runs)	1.002 (66)
24	1024	1.013	1.006
24	4096	1.011	0.996
24	16384	1.022	1.000
25	4096	1.009	1.002
25	16384	1.013	1.001

Both models hold to within about 2% [pcc:unverified], so the factor 2 per target is the whole story of the target phase.

Check 2: choosing the offsets cannot recover it (combinatorial simulation)

After phase 1 the IC knows every table log, so in principle it could choose its trial offsets w_k to avoid re-covering logs already covered, as BSGS's giant steps do. `translate_sim.py` measures the exact expected hit time E = sum_k (1 - |U_k|/n) for a random K-subset of Z_n (n = 2^18, 3 seeds) under three offset strategies: random; greedy (each step, the best of R random candidates by new coverage); and the tiled interval (BSGS). Data `translate_sim.json`; figure `gap.svg`/`gap.png`.

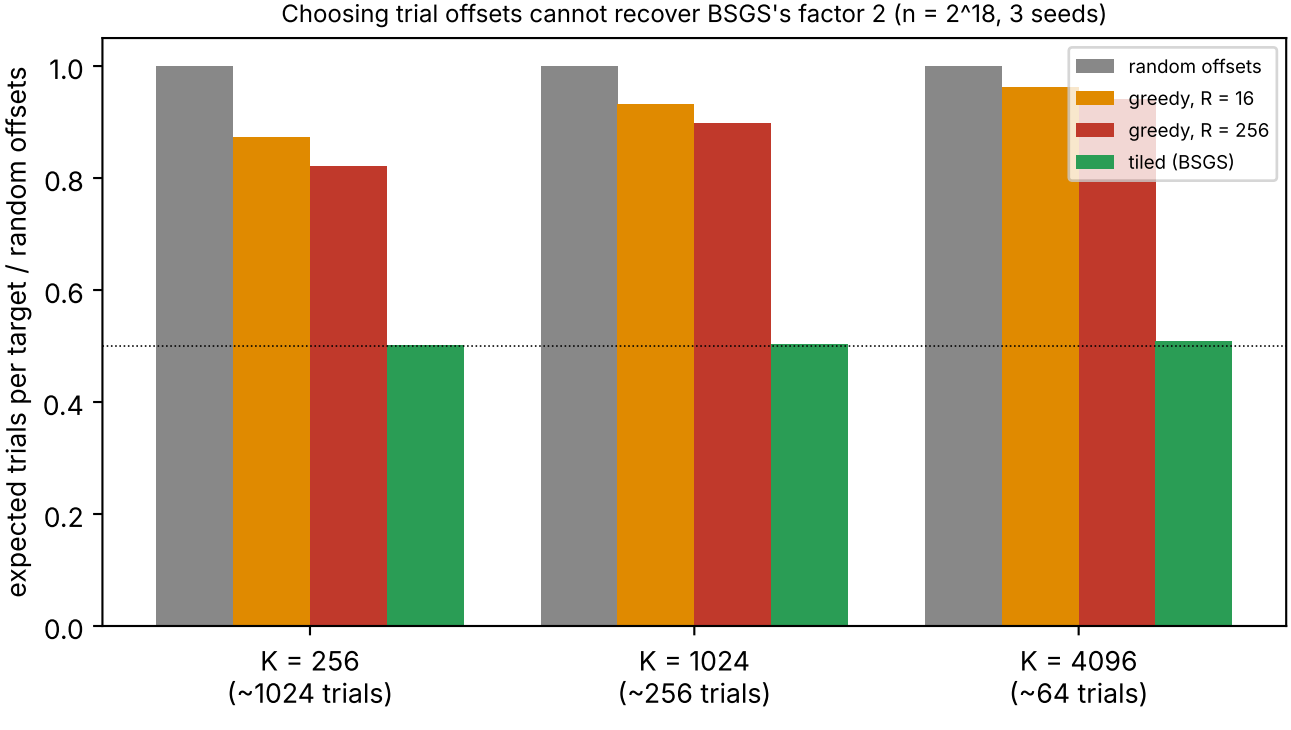
table K	trials (random)	greedy R = 16	greedy R = 256	tiled (BSGS)
256	1024	0.873	0.821	0.500
1024	256	0.930	0.896	0.500
4096	64	0.963	0.943	0.500

(columns 3 to 5: expected trials relative to random offsets; models n/K and (n/K + 1)/2 reproduce columns 2 and 5 exactly.) In the IC's regime (tens of trials per target) the best greedy choice saves 4% to 6%, not 50%, and computing it costs about R x K table lookups per step: at the §5a price of 1 GAE per unpriced operation that is far more than it saves. A random set's translates cannot be packed: once the union has density u, a new translate overlaps it in about 2Ku points whichever candidate is chosen.

Conclusion

The gap to BSGS comes from the table's structure, not from arithmetic or trial selection. Closing it needs a table whose logs form an interval (or a sum-set of arithmetic progressions), which requires a factor base with known logs: that is BSGS itself, a relabelling, and not an index-calculus improvement. Within generic algorithms the IC therefore cannot do better than tie BSGS, and with random-log tables it cannot even tie in the target phase. Beating BSGS (and rho by more than constants) needs a non-generic property of the curve, which none of these methods uses. Status: closed as a negative result for this family of methods [pcc:unverified].

Limits: the simulation is combinatorial (no curve arithmetic) on plain sets; greedy candidates are random, and a smarter search could do somewhat better, but the packing argument bounds any strategy far above the tiled 0.5 in this regime.


Choosing trial offsets cannot recover BSGS's factor 2 (n = 2^18, 3 seeds)
expected trials per target / random offsets
1.0
0.8
0.6
0.4
0.2
0.0
random offsets
greedy, R = 16
greedy, R = 256
tiled (BSGS)
K = 256
(~1024 trials)
K = 1024
(~256 trials)
K = 4096
(~64 trials)